County-Level Lung Cancer Mortality in the United States:

Social Vulnerability, Behavioral Risk Factors, Spatial Clustering, and Unexplained Mortality Hotspots

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A Comprehensive Project Report

Independent Applied Statistics and Public Health Data Analysis

Data sources: CDC Social Vulnerability Index, CDC PLACES, and NCI State Cancer Profiles

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Abstract

Lung cancer mortality varies sharply across U.S. counties, reflecting a combination of behavioral risk, social disadvantage, access to care, and geographic context. This project examined county-level lung cancer mortality using data from the CDC Social Vulnerability Index (SVI), CDC PLACES, and NCI State Cancer Profiles. The complete analytic sample contained 2,536 counties. Ordinary least squares regression with heteroskedasticity-robust standard errors was used to estimate associations between mortality and SVI, smoking, obesity, insurance status, and rurality. Quantile regression examined whether effects differed in high-mortality counties. Spatial dependence was assessed using Global Moran's I, Local Indicators of Spatial Association (LISA), and a spatial error model. Supplementary analyses used Random Forest and XGBoost with SHAP interpretation, a county-level COPD mediation analysis, and an SVI-by-rurality interaction.

Mean lung cancer mortality was 39.6 deaths per 100,000. COPD and smoking were the strongest bivariate correlates of mortality ($r = 0.721$ and $r = 0.712$, respectively), while overall SVI was much more weakly correlated ($r = 0.222$). In the primary adjusted model, smoking remained the dominant predictor ($\hat{\beta} = 2.23$, $p < 0.001$), whereas overall SVI was not independently significant. The smoking coefficient increased from 2.38 at the conditional median to 3.06 at the 90th percentile, showing a stronger association among the highest-mortality counties. Mortality was spatially clustered (Moran's $I = 0.542$, $p = 0.001$), and significant clustering remained in adjusted residuals ($I = 0.291$). LISA identified 232 High-High residual counties with mortality that remained unusually high after adjustment. The spatial error model improved fit substantially ($\Delta\text{AIC} = -508.5$, $\lambda = 0.491$, $p < 0.001$). A supplementary mediation analysis estimated that 50.8% of the county-level smoking-mortality association operated through COPD, and an SVI-by-rurality interaction suggested that social vulnerability had a stronger mortality association in rural counties.

Overall, the project shows that smoking and COPD account for much of the geographic variation in lung cancer mortality, but important unexplained spatial clusters remain. These counties may be priority areas for tobacco control, COPD management, lung cancer screening, and investigation of additional environmental and healthcare factors. Because the analysis is cross-sectional and ecological, results should be interpreted as county-level associations rather than individual causal effects.

1 Introduction

1.1 Background and Motivation

Lung cancer remains one of the most consequential causes of cancer mortality in the United States. National declines in smoking have reduced mortality over time, but the burden is not geographically uniform. Rural counties, Appalachia, parts of the South, and selected Midwestern areas continue to experience substantially higher death rates than many metropolitan areas (Islami et al., 2018; Lerro et al., 2020; Withrow et al., 2020).

Smoking is the most important established behavioral determinant of lung cancer, but county mortality patterns are also shaped by chronic respiratory disease, socioeconomic conditions, healthcare access, rurality, and other place-based factors. The CDC Social Vulnerability Index provides a multidimensional measure of socioeconomic, household, minority/language, and housing/transportation vulnerability (CDC/ATSDR, 2024). CDC PLACES adds local estimates of smoking, obesity, COPD, insurance status, physical inactivity, and other health indicators (CDC, 2025; Greenlund et al., 2022). When linked with mortality data from NCI State Cancer Profiles, these sources make it possible to examine both measured risk factors and residual geographic patterns.

The central practical question is not simply whether smoking predicts mortality. It is also whether counties remain unusually burdened after accounting for smoking and related characteristics. Such counties may indicate unmeasured environmental exposures, occupational hazards, screening gaps, healthcare limitations, or other structural conditions that conventional regression models do not capture.

1.2 Research Questions

This study addresses six questions:

1. How does lung cancer mortality vary across U.S. counties and rural/urban settings?
2. Which behavioral and social indicators are most strongly associated with mortality?
3. Does overall social vulnerability remain associated with mortality after adjustment for major behavioral and structural covariates?
4. Are predictor effects stronger in counties at the upper end of the mortality distribution?
5. Does substantial spatial clustering remain after regression adjustment, and where are the unexplained mortality hotspots?
6. Do COPD and rurality help explain or modify the relationships observed in the primary models?

1.3 Conceptual Framework

The analysis treats county lung cancer mortality as the result of overlapping behavioral, clinical, social, and spatial pathways. Smoking is expected to have the strongest direct association.

COPD is treated as a marker of cumulative respiratory injury and a possible intermediate pathway linking smoking burden with mortality. Social vulnerability may influence smoking exposure, prevention, screening, treatment access, and other health resources. Rurality may intensify these relationships through transportation barriers, provider shortages, and limited access to low-dose CT screening. Finally, spatially clustered residuals are interpreted as evidence that neighboring counties share additional unmeasured determinants.

2 Data and Methods

2.1 Data Sources and Analytic Sample

The analysis combined three primary sources. Lung cancer mortality came from NCI State Cancer Profiles (NCI, 2024). County social vulnerability measures came from the 2022 CDC/ATSDR Social Vulnerability Index (CDC/ATSDR, 2024). Behavioral and health measures came from CDC PLACES (CDC, 2025). County boundaries and contiguity information were derived from Census geographic files where needed for spatial analysis (U.S. Census Bureau, 2022).

The complete analytic sample contained 2,536 counties. Spatial analyses were restricted to 2,521 counties in the contiguous United States with valid geographic neighbors. Suppressed mortality values were concentrated in smaller counties and were excluded from the complete-case analytic sample.

2.2 Variables

The outcome was age-adjusted lung cancer mortality per 100,000 population. The main explanatory variables were overall SVI score, smoking prevalence, obesity prevalence, uninsured prevalence, and rural/urban status. Additional variables included COPD prevalence, physical inactivity, annual checkup prevalence, and the four SVI domain scores: socioeconomic status, household characteristics, minority/language, and housing/transportation.

2.3 Regression Analysis

The primary model was

$$Y_i = \beta_0 + \beta_1 SVI_i + \beta_2 Smoke_i + \beta_3 Obesity_i + \beta_4 Uninsured_i + \beta_5 Urban_i + \varepsilon_i, \tag{1}$$

where Y_i is county lung cancer mortality. HC1 heteroskedasticity-consistent standard errors were used (White, 1980). A second model replaced overall SVI with its four component themes to determine whether opposing domain-specific associations were hidden by the composite score.

2.4 Quantile Regression

Ordinary least squares estimates the conditional mean. To determine whether associations differed among high-mortality counties, quantile regression was estimated at $\tau = 0.50$, 0.75, and

0.90 (Koenker and Bassett, 1978). The same covariates used in the primary OLS model were retained.

2.5 Spatial Analysis

Global Moran's I was used to test spatial autocorrelation in raw mortality and regression residuals (Moran, 1950). Queen-contiguity spatial weights were row-standardized. Local Moran statistics were used to classify counties into High-High, Low-Low, High-Low, and Low-High clusters (Anselin, 1995). High-High residual clusters were interpreted as counties with higher-than-predicted mortality surrounded by counties showing the same pattern.

Because OLS residuals remained spatially dependent, a maximum-likelihood spatial error model was estimated:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{2}$$

$$\mathbf{u} = \lambda \mathbf{W}\mathbf{u} + \boldsymbol{\varepsilon}, \tag{3}$$

where $\mathbf{W}$ is the spatial-weights matrix and λ measures spatial dependence in the error process (Anselin, 1988; LeSage and Pace, 2009).

2.6 Supplementary Analyses

Random Forest and XGBoost classifiers were used as supplementary predictive analyses, with SHAP values used to summarize feature importance and direction (Breiman, 2001; Chen and Guestrin, 2016; Lundberg and Lee, 2017). A Baron-Kenny style county-level mediation analysis examined the pathway Smoking → COPD → Lung Cancer Mortality (Baron and Kenny, 1986). An interaction between SVI and rurality assessed whether the social vulnerability-mortality association differed between rural and urban counties.

3 Results

3.1 Descriptive Statistics

Lung cancer mortality averaged 39.59 deaths per 100,000 (SD 11.54), with values ranging from 9.1 to 99.0. Smoking prevalence averaged 16.45%, COPD 7.08%, and obesity 37.83%. Rural counties represented 57.6% of the sample and had higher unadjusted mortality than urban counties: 41.7 versus 36.7 deaths per 100,000.

Table 1: Selected descriptive statistics ($N = 2,536$ counties)

Variable	Mean	SD	Min	Median	Max
Lung mortality (per 100k)	39.59	11.54	9.1	39.0	99.0
SVI overall score	0.53	0.29	0.00	0.53	1.00
Smoking prevalence (%)	16.45	3.52	6.5	16.3	40.3
Obesity prevalence (%)	37.83	4.73	17.1	38.4	54.0
COPD prevalence (%)	7.08	1.57	3.2	6.9	15.0
Uninsured adults (%)	11.47	4.43	4.0	10.5	43.7
Physical inactivity (%)	27.20	5.35	12.2	26.7	48.9

3.2 Correlation Patterns

COPD and smoking were the variables most strongly associated with lung cancer mortality. Their Spearman correlations were 0.721 and 0.712, respectively. Physical inactivity and obesity showed moderate positive correlations. Overall SVI was positively but much more weakly related to mortality. The minority/language SVI theme was negatively correlated with mortality, illustrating why a single composite vulnerability score can conceal distinct underlying patterns.

Table 2: Spearman correlations with lung cancer mortality

Variable	Spearman r
COPD prevalence	0.721
Smoking prevalence	0.712
Physical inactivity	0.529
Obesity prevalence	0.516
SVI socioeconomic theme	0.328
SVI household theme	0.242
SVI overall	0.222
Uninsured adults	0.197
SVI minority/language theme	-0.178

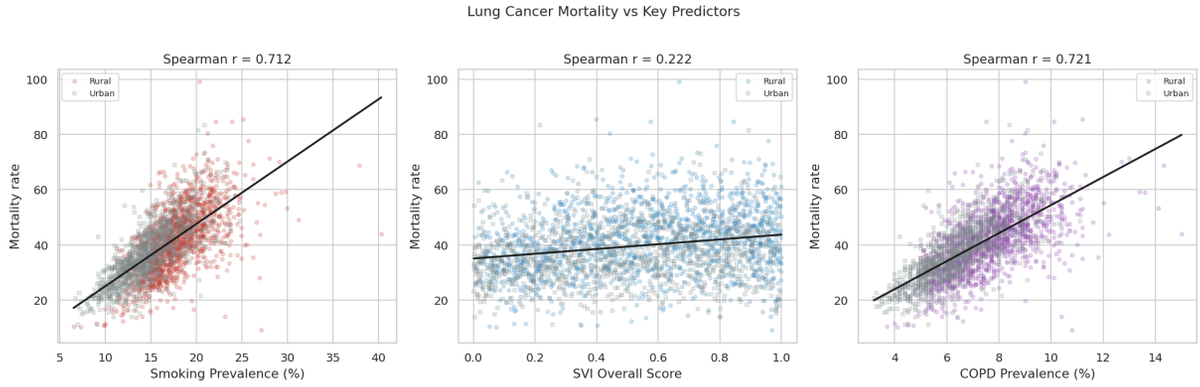


Figure 1: Bivariate relationships between lung cancer mortality and smoking, SVI, and COPD. Smoking and COPD show substantially stronger relationships with mortality than overall SVI.

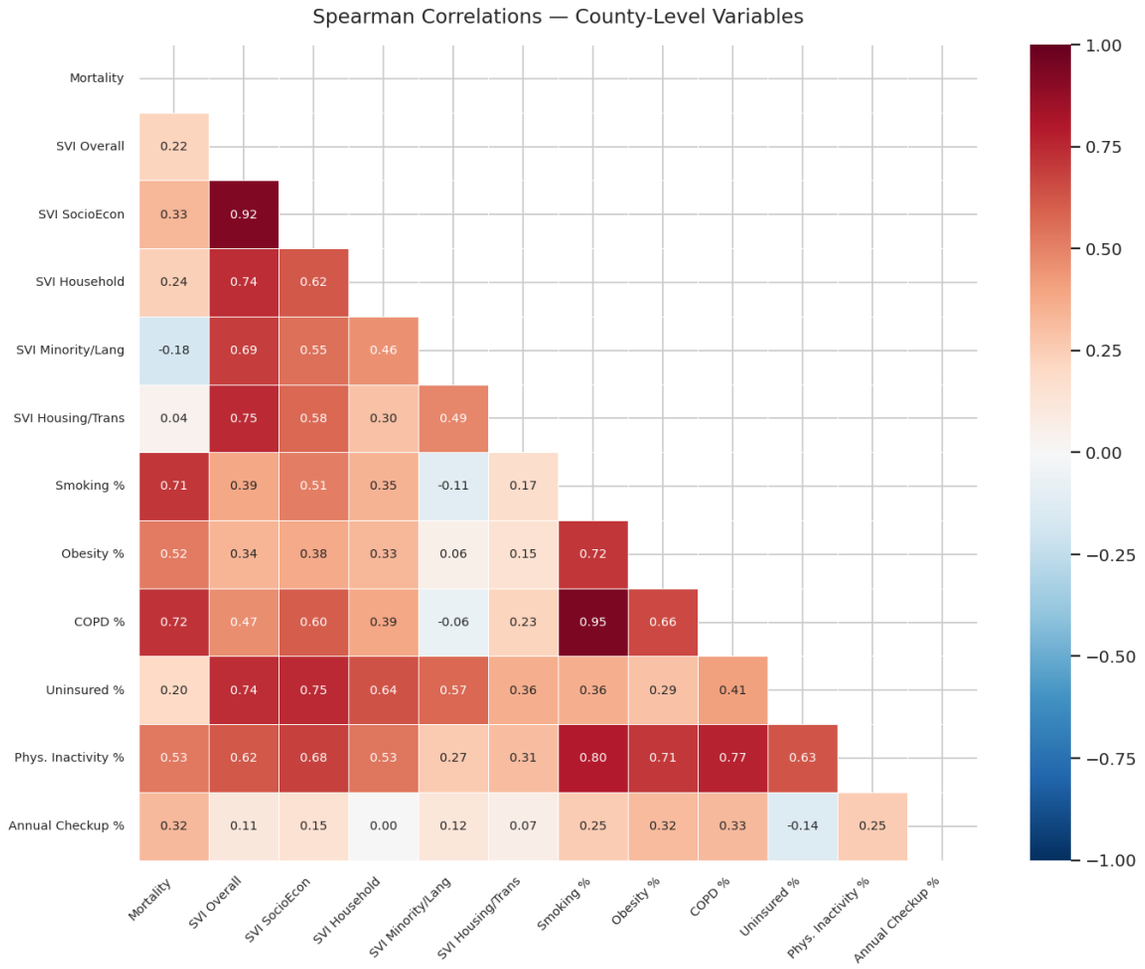


Figure 2: Spearman correlation heatmap for the main analytic variables.

3.3 Primary Regression Results

Adding smoking to an SVI-only model increased explained variance from about 5% to 48%, showing that behavioral exposure accounted for a large share of the between-county variation. The full primary model explained approximately half of the mortality variation (adjusted $R^2 = 0.499$).

Smoking had by far the largest adjusted coefficient. Each one-percentage-point increase in smoking prevalence was associated with an estimated 2.23 additional lung cancer deaths per 100,000, holding other model variables constant. Obesity also had a positive association. Overall SVI was not statistically significant after adjustment ($p = 0.278$).

Table 3: Primary OLS model with HC1 robust standard errors

Predictor	Estimate	SE	p-value	95% CI
SVI overall	0.88	0.81	0.278	[-0.71, 2.47]
Smoking prevalence	2.23	0.12	< 0.001	[2.00, 2.46]
Obesity prevalence	0.24	0.06	< 0.001	[0.13, 0.36]
Uninsured adults	-0.39	0.05	< 0.001	[-0.49, -0.30]
Urban (ref.=Rural)	2.04	0.37	< 0.001	[1.31, 2.77]

The negative uninsured coefficient and positive adjusted urban coefficient should not be read as causal effects. They arise in a multivariable ecological model containing highly structured geographic and behavioral covariates. The safer interpretation is that, conditional on smoking, obesity, SVI, and rurality, these variables capture residual between-county differences rather than direct mechanisms.

When overall SVI was replaced by its component domains, the socioeconomic and household themes were positively associated with mortality, whereas the minority/language and housing/transportation themes were negative. These opposing directions help explain why the composite SVI score had a weak adjusted association.

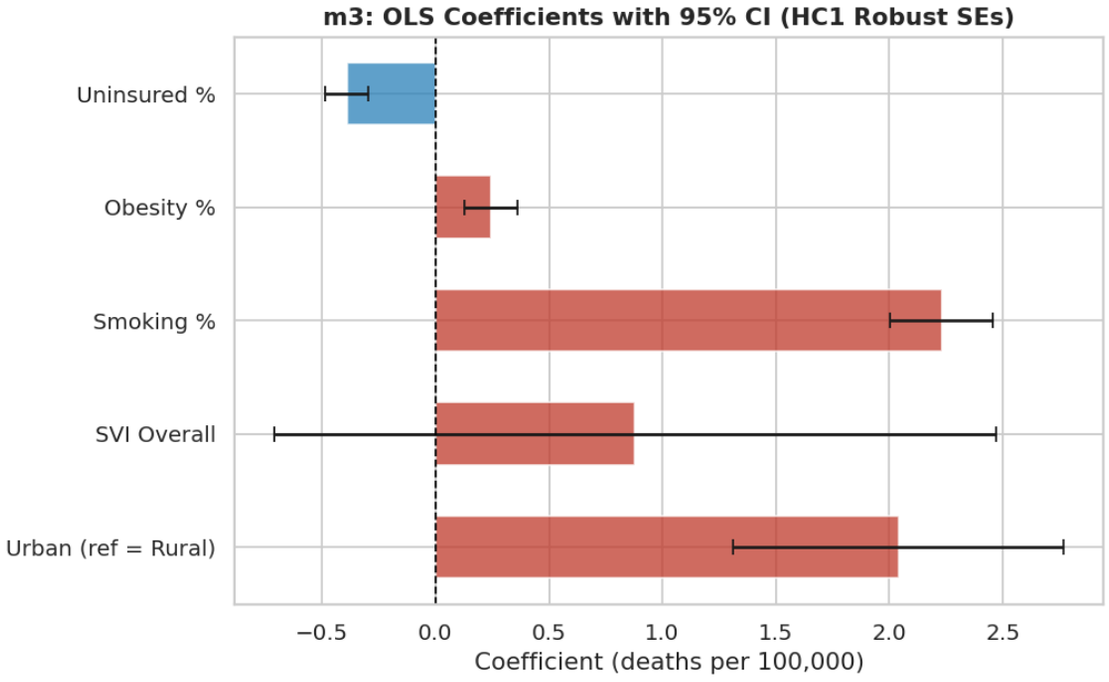


Figure 3: Coefficient plot for the primary adjusted OLS model.

3.4 Quantile Regression

The association between smoking and mortality became stronger at higher conditional mortality quantiles. The coefficient increased from 2.38 at the median to 2.80 at the 75th percentile and 3.06 at the 90th percentile. This result indicates that smoking is especially strongly associated with mortality in counties already experiencing the highest burden.

Table 4: OLS and quantile-regression coefficient comparison

Predictor	OLS	Q50	Q75	Q90
Smoking	2.23	2.38	2.80	3.06
SVI overall	0.88	1.98	-1.30	-4.10
Obesity	0.24	0.22	0.17	0.22
Uninsured	-0.39	-0.47	-0.30	-0.13
Urban	2.04	2.13	1.84	0.59

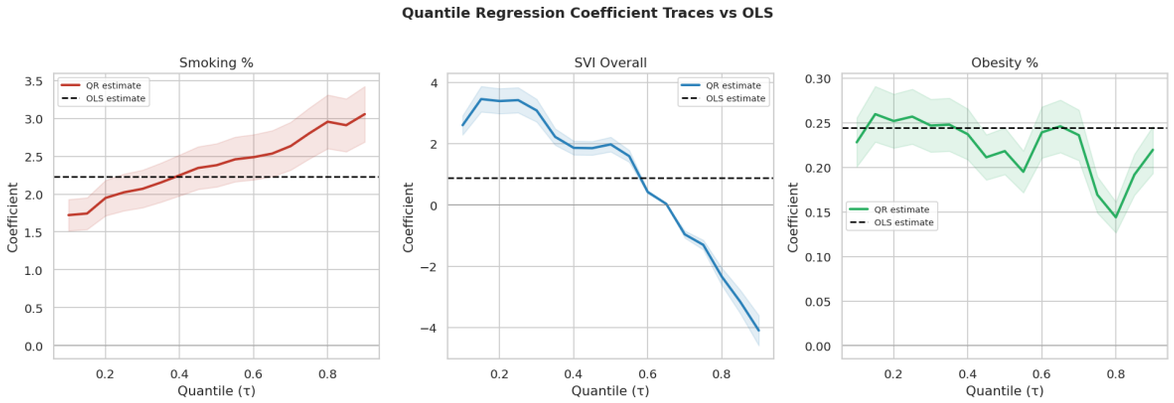


Figure 4: Quantile-regression coefficient traces. The smoking coefficient rises across the mortality distribution.

3.5 Geographic Patterns

The mortality and smoking maps showed strong geographic alignment, particularly across Appalachia, the Deep South, the lower Mississippi Valley, and parts of the Midwest. The SVI map overlapped only partially with the mortality map, supporting the regression finding that overall social vulnerability alone does not summarize the dominant geographic risk pattern.

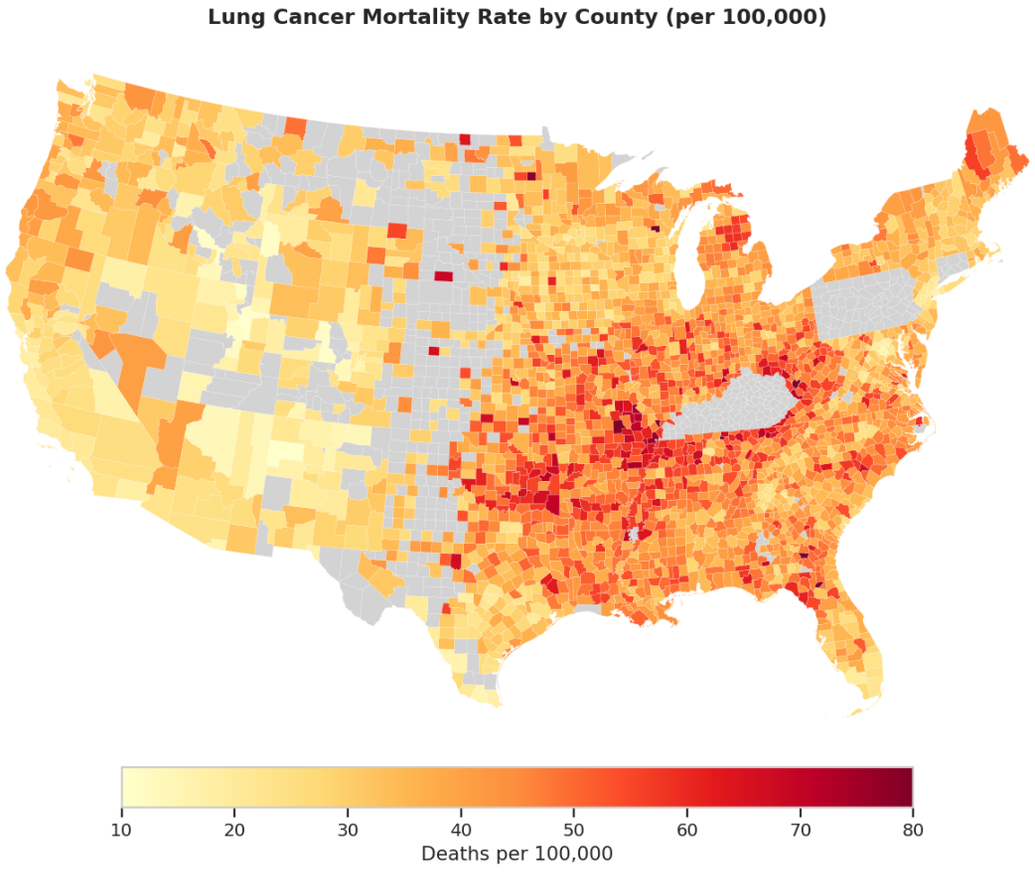


Figure 5: County-level lung cancer mortality. High-burden areas are concentrated in Appalachia, the South, and selected Midwestern regions.

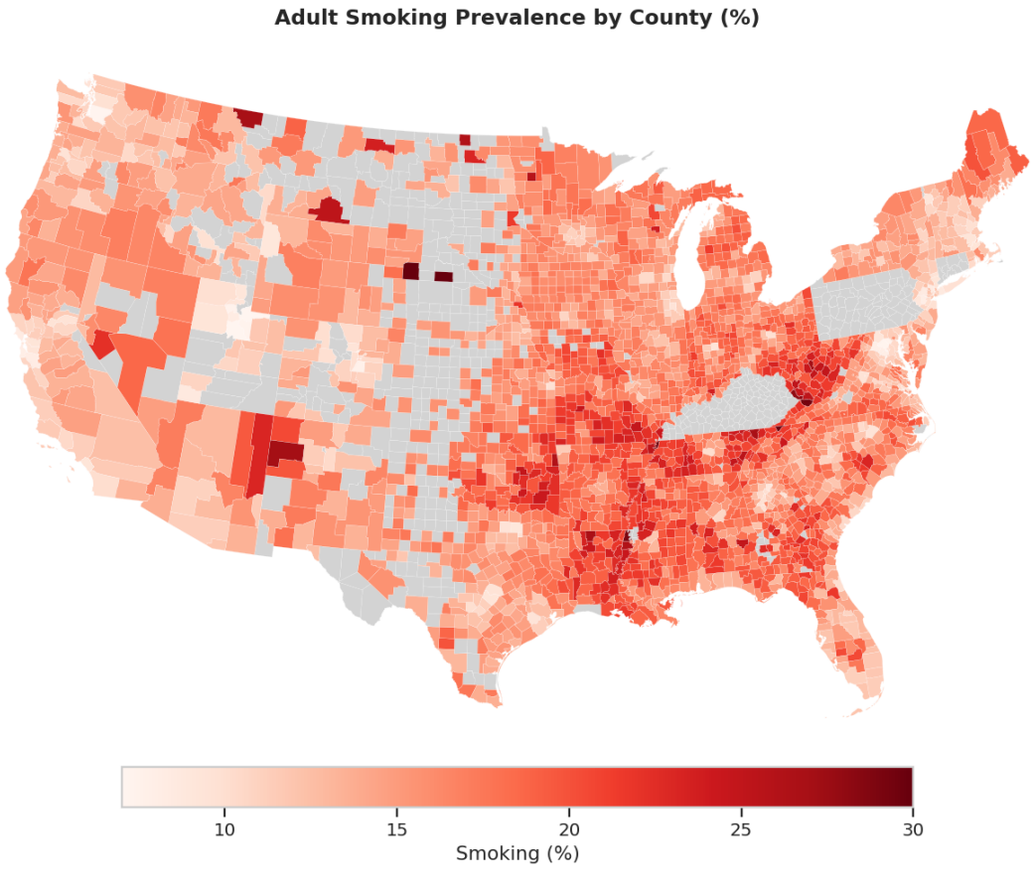


Figure 6: County-level smoking prevalence. The geographic pattern closely resembles the mortality map.

3.6 Spatial Autocorrelation and Residual Hotspots

Raw lung cancer mortality was strongly spatially clustered (Moran's $I = 0.542$, $p = 0.001$). After adjustment using the primary OLS model, residual clustering remained substantial ($I = 0.291$, $p = 0.001$). This means that counties with unexpectedly high or low mortality tended to be geographically near counties with similar unexplained deviations.

LISA identified 232 High-High residual counties, representing 9.2% of the 2,521-county spatial sample. These counties averaged 50.19 deaths per 100,000 compared with 38.54 among other counties. Their measured risk factors were only modestly higher, indicating that the mortality excess was not fully explained by the variables in the primary model. High-High clusters were especially common in Illinois, Texas, Arkansas, Tennessee, Oklahoma, Missouri, and Florida.

Table 5: LISA classification of OLS residuals

Cluster	Counties	Percent
Not significant	1,887	74.9%
High-High	232	9.2%
Low-Low	275	10.9%
Low-High	87	3.5%
High-Low	40	1.6%

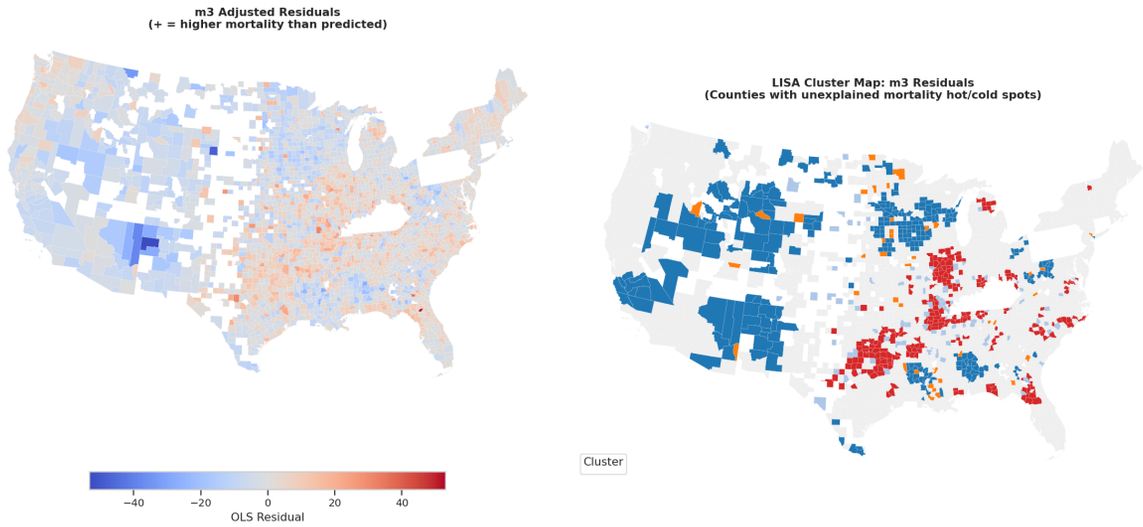


Figure 7: OLS residual map and LISA cluster map. High-High counties represent geographically concentrated mortality that remains above model predictions.

3.7 Spatial Error Model

The spatial error model improved fit sharply relative to OLS: AIC decreased from 17,858 to 17,350, a reduction of 508.5 points. The spatial dependence parameter was positive and highly significant ($\lambda = 0.491$, $p < 0.001$), confirming that neighboring counties shared unmeasured characteristics. Smoking remained robust in the spatial model, with only modest attenuation from 2.23 to 2.07. Overall SVI remained non-significant.

Table 6: OLS versus spatial error model

Predictor	OLS	SEM	SEM p-value
SVI overall	0.88	-0.57	0.500
Smoking	2.23	2.07	< 0.001
Obesity	0.24	0.19	< 0.001
Uninsured	-0.39	-0.37	< 0.001
Spatial error λ	–	0.491	< 0.001
AIC	17,858	17,350	–

3.8 Supplementary Analyses

Random Forest and XGBoost both achieved an AUC of 0.852 in classifying high-mortality counties. Feature-importance and SHAP summaries consistently ranked COPD and smoking among the most influential predictors. This reinforces the central result from the regression analyses rather than replacing it.

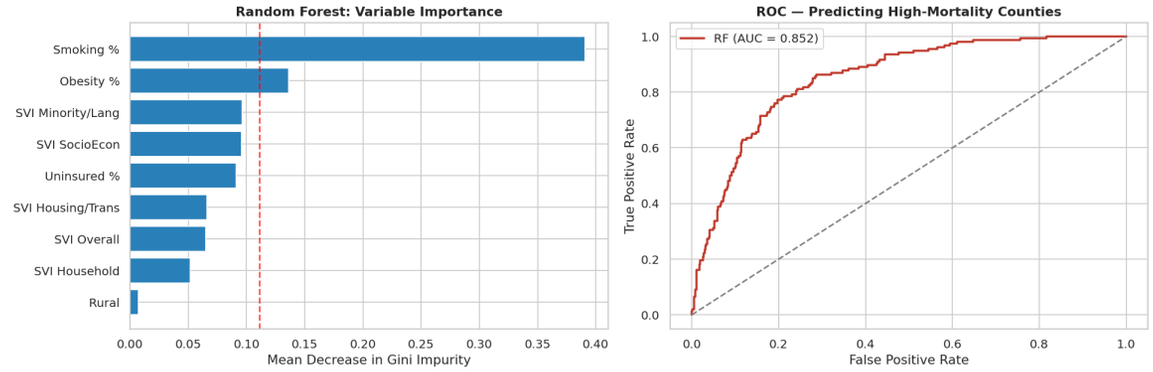


Figure 8: Random Forest variable importance and ROC performance.

The county-level mediation analysis estimated an indirect smoking effect through COPD of 1.145, compared with a total effect of 2.255. This corresponds to 50.8% mediated. Because the data are cross-sectional and ecological, this should be interpreted as a descriptive decomposition consistent with a plausible pathway, not as proof of individual causal mediation.

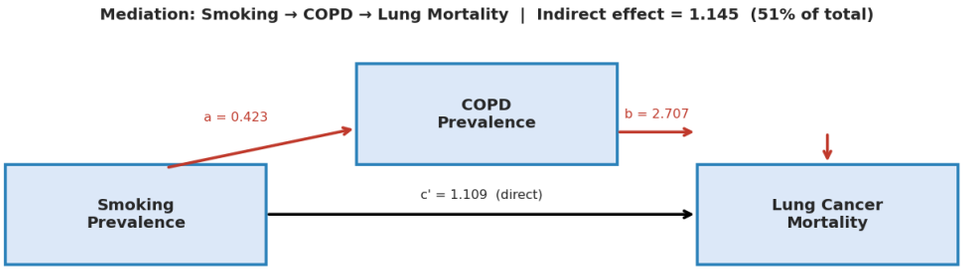


Figure 9: Smoking-COPD-mortality mediation diagram. Approximately 50.8% of the county-level smoking association was statistically mediated through COPD.

The SVI-by-rurality interaction was statistically significant ($p = 0.031$). In rural counties, higher SVI was positively associated with mortality, whereas the estimated SVI slope in urban counties was close to zero. This suggests that the consequences of social vulnerability may depend on geographic context.

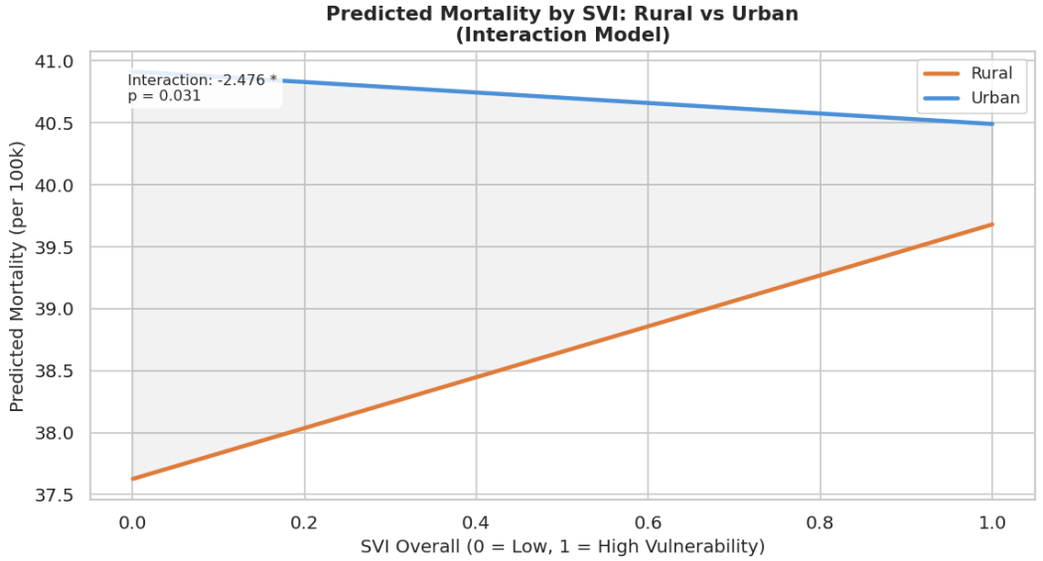


Figure 10: Marginal association between SVI and mortality by rural/urban status.

4 Discussion

4.1 Main Findings

Four findings organize the project. First, smoking and COPD were the strongest and most consistent county-level markers of lung cancer mortality. Smoking alone explained a large increase in model fit, and its association remained strong after adjustment and in the spatial error model. Second, smoking was even more strongly associated with mortality among counties at the upper end of the mortality distribution. Third, overall SVI was much less informative than its component domains and did not remain independently significant after behavioral adjustment. Fourth, substantial geographic clustering persisted after conventional regression, producing 232 High-High residual counties with mortality well above predicted levels.

These findings suggest that the national county-level burden is not adequately summarized by a simple vulnerable-versus-not-vulnerable distinction. Behavioral exposure, especially smoking, dominates the measured signal, while social vulnerability appears to operate in a more heterogeneous and context-dependent way. The SVI theme decomposition and rural interaction both support this interpretation.

4.2 Interpretation of Spatial Hotspots

The residual hotspot analysis is particularly important because it separates high observed mortality from high mortality that is already explained by measured covariates. A county can have high mortality because smoking prevalence is high, but a High-High residual county has mortality that remains high even after the model accounts for smoking, obesity, SVI, insurance status, and rurality. Clusters of these counties may reflect shared but unmeasured exposures or healthcare conditions.

Potential explanations include radon, occupational and industrial carcinogen exposure, historical tobacco intensity not captured by current smoking prevalence, delayed diagnosis, limited access

to thoracic oncology services, differences in screening uptake, and other socioeconomic or environmental factors. These mechanisms were not measured directly in the present dataset and therefore should be treated as hypotheses for follow-up rather than conclusions.

4.3 Public Health Implications

The strongest immediate implication is the continuing importance of tobacco control. Counties at the highest end of the mortality distribution appear particularly sensitive to smoking prevalence, making cessation programs, taxation, smoke-free policies, and prevention efforts especially relevant in high-burden regions.

The COPD results also support closer integration of respiratory disease management and lung cancer prevention. Counties with high smoking and COPD prevalence may benefit from combined tobacco cessation, COPD case finding, and low-dose CT screening outreach. The 232 residual High-High counties provide a second targeting framework: they identify places where standard risk profiles do not fully account for mortality and where deeper local investigation may be justified.

Finally, the rural SVI interaction suggests that the same level of social vulnerability may have different consequences depending on healthcare geography. Rural areas may require strategies that address travel distance, workforce shortages, referral pathways, and mobile or regionalized screening services rather than simply expanding programs designed for metropolitan settings.

4.4 Strengths and Limitations

The project combines several complementary methods rather than relying on a single regression model. Robust OLS estimates, quantile regression, spatial diagnostics, a spatial error model, machine learning, and supplementary mediation and interaction analyses all point to related features of the same county-level pattern. The use of national data also supports broad geographic comparisons.

The analysis nevertheless has important limitations. It is cross-sectional, so temporal ordering cannot be established. The unit of analysis is the county, creating a risk of ecological fallacy if results are interpreted at the individual level. PLACES variables are model-based small-area estimates rather than direct measurements in every county. Mortality suppression creates missingness in smaller counties. Current smoking prevalence may not represent historical exposure, which is especially important for a disease with long latency. Finally, the persistence of spatial autocorrelation after the SEM shows that relevant determinants remain unmeasured.

4.5 Future Work

Future versions should incorporate time-varying county data, historical smoking measures, low-dose CT screening uptake, healthcare capacity, radon exposure, occupational and industrial exposures, poverty and education, and rural-urban commuting classifications. A longitudinal spatial model could distinguish persistent high-risk counties from areas whose mortality is improving or worsening. Local spatial models such as geographically weighted regression or Bayesian disease-mapping approaches could also quantify regional heterogeneity more directly.

5 Conclusion

County-level lung cancer mortality in the United States is strongly structured by smoking, COPD, and geography. Smoking was the dominant measured predictor, remained robust under spatial adjustment, and became more influential at the highest mortality quantiles. Overall social vulnerability was comparatively weak after behavioral adjustment, but its component domains and rural interaction revealed important heterogeneity that a single composite score could not capture.

The spatial analyses add a practical contribution. Two hundred thirty-two counties formed High-High clusters of unexplained mortality, with death rates substantially above those of other counties even after adjustment. These areas are plausible priorities for intensified tobacco control, COPD management, lung cancer screening, and investigation of additional environmental and healthcare factors. The project therefore provides both an explanatory framework for national county variation and a geographically targeted strategy for identifying communities where conventional risk factors do not fully explain the observed burden.

Appendix: Additional Notes

Interpretation of coefficients. All regression coefficients are county-level conditional associations. They should not be interpreted as individual causal effects. In particular, counterintuitive signs on uninsured status, urbanicity, or individual SVI domains may reflect geographic structure, compositional differences, and adjustment for correlated predictors.

Spatial weights. Queen contiguity defines counties as neighbors when their boundaries or vertices touch. Weights were row-standardized before computing Moran's I, LISA statistics, and the spatial error model.

Machine learning. The Random Forest and XGBoost analyses are supplementary predictive checks. Their purpose is to assess whether nonlinear prediction ranks the same variables as the inferential models; they are not substitutes for the regression and spatial analyses.

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